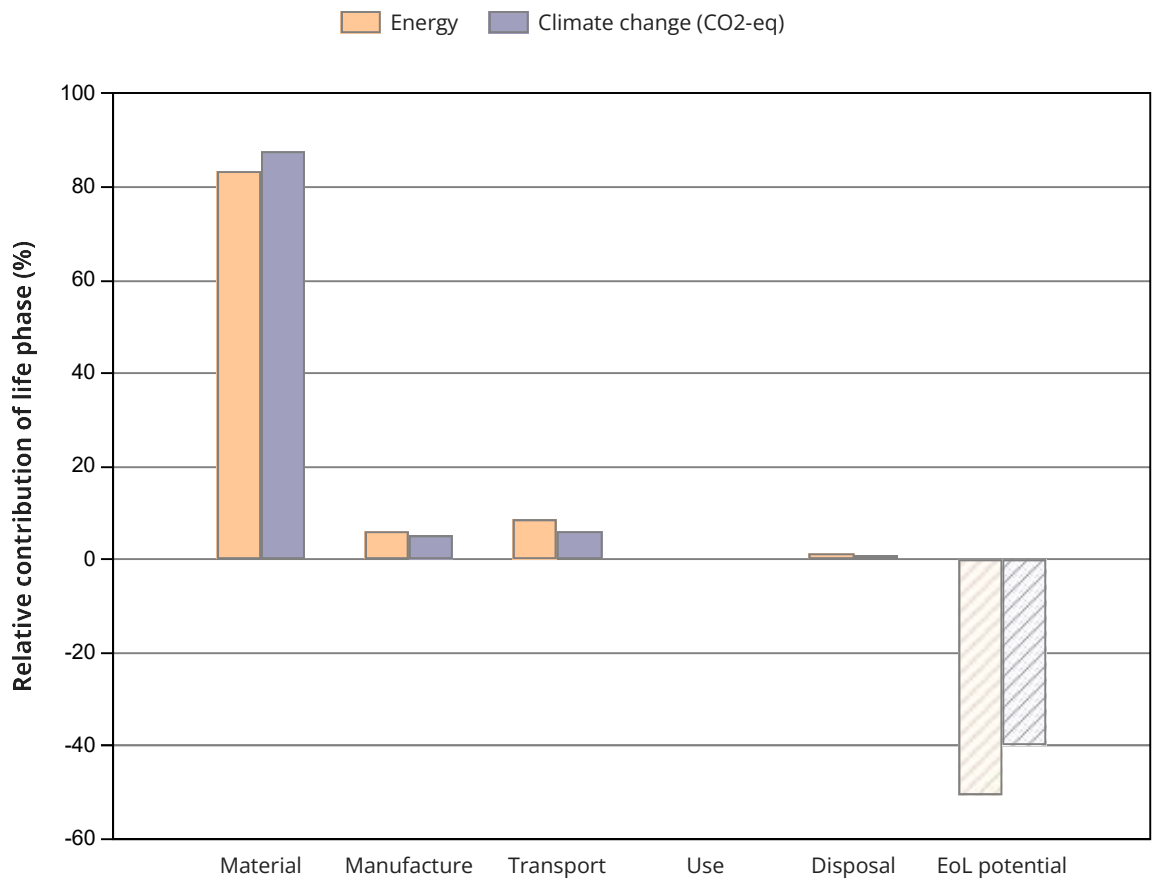
Eco Audit Report

Product name	LC3 Reactor
Country of use	Northern America
Product life (years)	30

Summary:



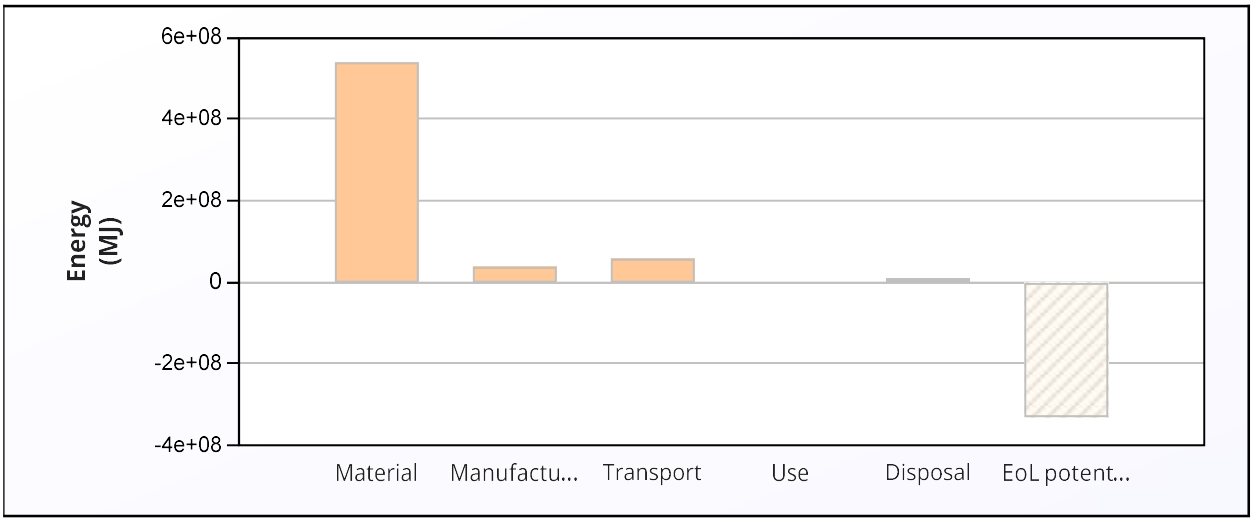
Energy details Climate change (CO2-eq) details

Phase	Energy (MJ)	Energy (%)	Climate change (CO2-eq) (kg)	Climate change (CO2-eq) (%)
Material	5.4e+08	83.5	5.14e+07	87.7
Manufacture	4.04e+07	6.3	3.03e+06	5.2
Transport	5.61e+07	8.7	3.49e+06	6.0
Use	0	0.0	0	0.0
Disposal	9.8e+06	1.5	6.86e+05	1.2
Total (for first life)	**6.46e+08**	**100**	**5.86e+07**	**100**
End of life potential	-3.28e+08		-2.34e+07	

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ENERGY ANALYSIS

Summary



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 30 year product life):	2.15e+07

DETAILED BREAKDOWN OF INDIVIDUAL LIFE PHASES

MATERIAL:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Concrete	Concrete Base***	Virgin (0%)	7.2e+07	1	7.2e+07	6.5e+07	12.0
Structural Steel	Low carbon steel	Virgin (0%)	6e+06	1	6e+06	1.9e+08	35.4
Modular Steel	Low carbon steel	Virgin (0%)	8e+06	1	8e+06	2.5e+08	47.2
LC3	Concrete LC3***	Virgin (0%)	4.8e+07	1	4.8e+07	2.9e+07	5.4
Total				**4**	**1.3e+08**	**5.4e+08**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

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MANUFACTURE:

Summary

Component	Process	Amount processed	Energy (MJ)	%
Structural Steel	Roll forming	6e+06 kg	1.7e+07	42.9
Modular Steel	Roll forming	8e+06 kg	2.3e+07	57.1
Total			**4e+07**	**100**

TRANSPORT:

Summary

BREAKDOWN BY TRANSPORT STAGE

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Concrete	Truck 16-32t, EURO 6	30	1.2e+07	20.7
Steel	Truck >32t, EURO 6	2e+02	4.4e+07	79.3
Total		**2.3e+02**	**5.6e+07**	**100**

BREAKDOWN BY COMPONENTS

Component	Mass (kg)	Energy (MJ)	%
Concrete	7.2e+07	3e+07	53.7
Structural Steel	6e+06	2.5e+06	4.5
Modular Steel	8e+06	3.3e+06	6.0
LC3	4.8e+07	2e+07	35.8
Total	**1.3e+08**	**5.6e+07**	**100**

USE:

Summary

RELATIVE CONTRIBUTION OF STATIC AND MOBILE MODES

Mode	Energy (MJ)	%
Static	0	
Mobile	0	
Total	**0**	**100**

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Summary

Disposal:

Component	End of life option	Energy (MJ)	%
Concrete	None	0	0.0
Structural Steel	Recycle	4.2e+06	42.9
Modular Steel	Recycle	5.6e+06	57.1
LC3	None	0	0.0
Total		**9.8e+06**	**100**

EoL potential:

Component	End of life option	Energy (MJ)	%
Concrete	None	0	0.0
Structural Steel	Recycle	-1.4e+08	42.9
Modular Steel	Recycle	-1.9e+08	57.1
LC3	None	0	0.0
Total		**-3.3e+08**	**100**

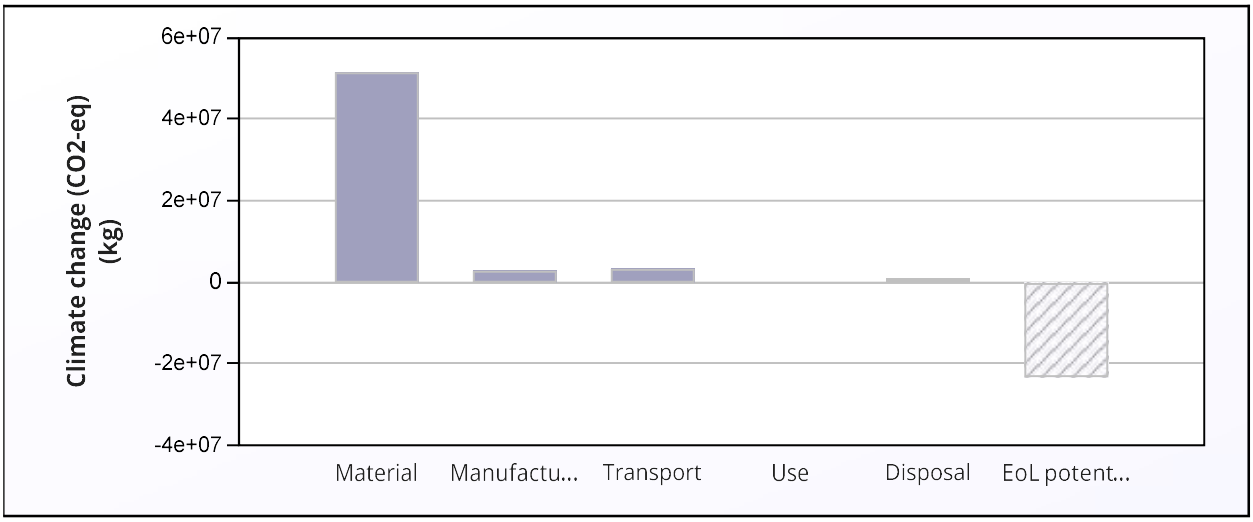
Summary

Notes:

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CLIMATE CHANGE (CO2-EQ) ANALYSIS

Summary



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 30 year product life):	1.95e+06

DETAILED BREAKDOWN OF INDIVIDUAL LIFE PHASES

MATERIAL:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Climate change (CO2-eq) (kg)	%
Concrete	Concrete Base***	Virgin (0%)	7.2e+07	1	7.2e+07	1.1e+07	22.2
Structural Steel	Low carbon steel	Virgin (0%)	6e+06	1	6e+06	1.4e+07	27.2
Modular Steel	Low carbon steel	Virgin (0%)	8e+06	1	8e+06	1.9e+07	36.3
LC3	Concrete LC3***	Virgin (0%)	4.8e+07	1	4.8e+07	7.4e+06	14.3
Total				**4**	**1.3e+08**	**5.1e+07**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

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MANUFACTURE:

Summary

Component	Process	Amount processed	Climate change (CO2-eq) (kg)	%
Structural Steel	Roll forming	6e+06 kg	1.3e+06	42.9
Modular Steel	Roll forming	8e+06 kg	1.7e+06	57.1
Total			**3e+06**	**100**

TRANSPORT:

Summary

BREAKDOWN BY TRANSPORT STAGE

Stage name	Transport type	Distance (km)	Climate change (CO2-eq) (kg)	%
Concrete	Truck 16-32t, EURO 6	30	7.6e+05	21.7
Steel	Truck >32t, EURO 6	2e+02	2.7e+06	78.3
Total		**2.3e+02**	**3.5e+06**	**100**

BREAKDOWN BY COMPONENTS

Component	Mass (kg)	Climate change (CO2-eq) (kg)	%
Concrete	7.2e+07	1.9e+06	53.7
Structural Steel	6e+06	1.6e+05	4.5
Modular Steel	8e+06	2.1e+05	6.0
LC3	4.8e+07	1.3e+06	35.8
Total	**1.3e+08**	**3.5e+06**	**100**

USE:

Summary

RELATIVE CONTRIBUTION OF STATIC AND MOBILE MODES

Mode	Climate change (CO2-eq) (kg)	%
Static	0	
Mobile	0	
Total	**0**	**100**

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Summary

Disposal:

Component	End of life option	Climate change (CO2-eq) (kg)	%
Concrete	None	0	0.0
Structural Steel	Recycle	2.9e+05	42.9
Modular Steel	Recycle	3.9e+05	57.1
LC3	None	0	0.0
Total		**6.9e+05**	**100**

EoL potential:

Component	End of life option	Climate change (CO2-eq) (kg)	%
Concrete	None	0	0.0
Structural Steel	Recycle	-1e+07	42.9
Modular Steel	Recycle	-1.3e+07	57.1
LC3	None	0	0.0
Total		**-2.3e+07**	**100**

Summary

Notes:

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Appendix

User-defined materials:

Material	Generic material class	Price (AUD)	Embodied energy, primary prod. (virgin grade) (MJ/kg)	Climate change (CO2-eq), primary production (virgin grade) (kg/kg)	Recycle	Embodied energy, recycling (MJ/kg)	Climate change (CO2-eq), recycling (kg/kg)
Concrete Base	Other	0.14	0.9	0.1583		0.0348	0.00313
Concrete LC3	Other	0.15	0.603	0.1533		0.0348	0.00313

Material	% recycled current supply	Recycle	Combust for energy recovery	Heat of combustion (net) (MJ/kg)	Combustion CO2 (kg/kg)	Landfill
Concrete Base		yes				
Concrete LC3		yes				yes

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